

skill-8

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jaswanth_reddy_ch@JERSEY:~$ mkdir skill-8.c
jaswanth_reddy_ch@JERSEY:~$ cd skill-8.c
jaswanth_reddy_ch@JERSEY:~/skill-8.c$ gedit skill-8.c
jaswanth_reddy_ch@JERSEY:~/skill-8.c$ gcc skill-8.c -o skill-8
jaswanth_reddy_ch@JERSEY:~/skill-8.c$ ./skill-8
=== Variable Expansion Engine ===
```

[Test 1: Simple Variable Reference]

Input : Hello \$USER, welcome!

Output: Hello alice, welcome!

[Test 2: Braced Variable Reference]

Input : Home dir is \${HOME}/bin

Output:

[Test 3: Undefined Variable]

Input : Value is: '\$UNDEFINED_VAR'

Output: Value is: ''

[Test 4: Nested Variable Reference]

Input : Key is: \${KEY_\${ROLE}}

Output:

[Test 5: Complex Multi-Nested Input]

Input : User \$USER (\${KEY_\${INNER}}) resides at \${HOME}

Output: